CEBALID Syndrome (MN1 C-Terminal Truncation Syndrome): A Comprehensive Disease Characterization

Disease: CEBALID Syndrome | **Synonyms:** MN1 C-Terminal Truncation (MCTT) Syndrome; MCTT **Causal gene:** *MN1* (Meningioma-1 / meningioma translocation 1), 22q12.1 | **MONDO:** MONDO:0032908 | **OMIM:** #618774 **Inheritance:** Autosomal dominant (de novo) | **Category:** Mendelian, rare neurodevelopmental/craniofacial malformation syndrome

Summary

CEBALID syndrome is an ultra-rare, congenital, autosomal-dominant neurodevelopmental and craniofacial malformation disorder caused by de novo C-terminal truncating variants in *MN1* (Meningioma-1), a two-exon transcriptional co-regulator gene on chromosome 22q12.1. The acronym CEBALID captures the core clinical domains: **C**raniofacial defects, dysmorphic **E**ars, **B**rain **A**bnormalities, **L**anguage delay, and **I**ntellectual **D**isability. The disorder was first delineated in 2020 through two landmark reports and has fewer than ~30 published cases worldwide as of 2025, so reliable population-level epidemiology is unavailable.

The defining molecular lesion is a truncating (nonsense or frameshift) *MN1* variant located in the terminal exon or the extreme 3′ end of exon 1. Because of this position, the mutant transcript **escapes nonsense-mediated mRNA decay (NMD)** and produces a stabilized, aggregation-prone, C-terminally truncated protein. Functional studies demonstrate this acts through a **gain-of-function / dominant change-of-function mechanism** rather than haploinsufficiency—a critical distinction, because whole-gene *MN1* deletions produce a milder, different phenotype. A recurrent hotspot variant, **p.Arg1295***, accounts for roughly one-third of reported probands.

The recognizable clinical gestalt is **midface hypoplasia** combined with **severe expressive language delay and intellectual disability**, dysmorphic/low-set ears, and a distinctive brain malformation—**partial (atypical) rhombencephalosynapsis**—frequently accompanied by

perisylvian polymicrogyria, corpus callosum anomalies, and craniofacial skeletal abnormalities. Mechanistically, MN1 functions as a **retinoic-acid-receptor (RAR/RXR) transcriptional coactivator** (working with p160/p300 coactivators) and acts **upstream of *Tbx22*** in posterior palate development; mouse *Mn1* knockouts die of cleft palate with agenesis of membranous cranial bones, linking MN1 dysfunction directly to the craniofacial phenotype. There is currently no disease-specific or curative therapy; diagnosis is by whole-exome/whole-genome sequencing plus characteristic brain MRI, and management is supportive and multidisciplinary.

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1. Disease Information

CEBALID syndrome (also termed **MN1 C-terminal truncation [MCTT] syndrome**) is a Mendelian, congenital neurodevelopmental and craniofacial malformation disorder. It was independently described in 2020 by Mak et al. (*Brain*) and Miyake et al. (*American Journal of Human Genetics*), who both identified clusters of individuals with de novo C-terminal truncating *MN1* variants sharing a remarkably consistent phenotype.

Key identifiers:

Resource	Identifier
MONDO	MONDO:0032908
OMIM	#618774
Gene	*MN1* (HGNC:7180), chromosome 22q12.1
Synonyms	MN1 C-terminal truncation syndrome; MCTT; CEBALID syndrome

As Dirani et al. (2025) state, *"CEBALID syndrome (Craniofacial defects, dysmorphic Ears, Brain Abnormalities, Language delay, and Intellectual Disability) is caused by C-terminal truncated transcriptional activator MN1"* and *"This novel syndrome ... was recently identified in 2020. Since then, less than 30 cases have been reported worldwide"* (PMID: 40713540).

Information source: The knowledge base is derived from aggregated, disease-level resources—primarily small case series and individual case reports (Mak n=23; Miyake n=3; Dirani n=1) combined with mechanistic model-organism and in vitro studies. There is no EHR/registry-scale dataset.

2. Etiology

Disease causal factor — genetic (de novo, dominant). CEBALID is a monogenic disorder caused by heterozygous **truncating variants in *MN1***. Mak et al. identified *"a cluster of de novo truncating mutations in MN1 in a cohort of 23 individuals"* (PMID: 31834374), and Miyake et al. described *"three individuals who have de novo MN1 variants that lead to a protein lacking the carboxyl (C) terminus"* (PMID: 31839203). The variants arise **de novo**, so affected individuals typically have unaffected parents.

Genetic risk factors. The sole established genetic risk factor is the presence of a de novo C-terminal truncating *MN1* variant. The recurrent **p.Arg1295*** variant represents a CpG-associated mutational hotspot rather than a founder allele. No susceptibility loci or modifier genes have been identified.

Environmental risk factors / protective factors / gene–environment interactions. None are established. As a de novo dominant malformation syndrome, there are **no known environmental risk factors, protective factors, lifestyle contributors, or gene–environment interactions**. Advanced paternal age is a general (not disease-specific) consideration for de novo variants but has not been formally studied here.

3. Phenotypes

The phenotype is **congenital/neonatal in onset**, **variable in severity but generally severe**, and **non-progressive (malformative/static)** with lifelong disability. Frequencies below derive chiefly from the Mak cohort (n=23) and the Miyake series (n=3).

Neurodevelopmental phenotypes

- **Intellectual disability** (HP:0001249) — core, high frequency
- **Severe expressive/speech language delay** (HP:0002474, HP:0000750) — hallmark, nearly universal
- **Global developmental delay** (HP:0001263)
- **Hypotonia** (HP:0001252)

Mak et al.: *"strikingly similar dysmorphic facial features, especially midface hypoplasia, and intellectual disability with severe expressive language delay"* (PMID: 31834374).

Brain malformation phenotypes

- **Partial/atypical rhombencephalosynapsis** (cerebellar vermis hypoplasia/aplasia with hemispheric fusion) — **8/10 imaged individuals** in Mak et al.
- **Perisylvian polymicrogyria** (HP:0032420 / HP:0002126)
- **Corpus callosum anomalies** including agenesis of the genu (HP:0001273)
- **Ventriculomegaly** (HP:0002119)
- **Abnormal posterior clinoid processes; persistent trigeminal artery**

Mak et al. identified *"an atypical form of rhombencephalosynapsis, a distinctive brain malformation characterized by partial or complete loss of the cerebellar vermis with fusion of the cerebellar hemispheres, in 8/10 individuals"* (PMID: 31834374). Dirani et al. documented a neonate whose MRI *"revealed polymicrogyri, agenesis of the genu of the corpus callosum and rhombencephalosynapsis"* (PMID: 40713540).

Craniofacial phenotypes

- **Midface hypoplasia** (HP:0011800) — hallmark gestalt feature
- **Hypertelorism** (HP:0000316); **downslanting palpebral fissures** (HP:0000494)
- **Proptosis / shallow orbits** (HP:0000520)
- **Depressed/broad nasal bridge** (HP:0005280)
- **Small, dysmorphic, low-set ears** (HP:0000369, HP:0000368)
- **High-arched or cleft palate** (HP:0000218 / HP:0000175)
- **Craniosynostosis and abnormal skull shape** (HP:0001363)

Other systemic phenotypes

- **Conductive and/or sensorineural hearing loss** (HP:0000405 / HP:0000407)
- **Feeding difficulties** (HP:0011968)
- **Multiple joint contractures / arthrogryposis** (HP:0002804) — severe neonatal presentation
- **Poor cardiopulmonary adaptation** in severe neonatal cases

Dirani et al. described a severe neonatal case in which *"the patient had poor extra-uterine cardiopulmonary adaptation that required mechanical ventilation, and feeding difficulties were noted with hypotonicity"* (PMID: 40713540), expanding the phenotype to include congenital arthrogryposis.

Quality-of-life impact: Severe expressive language impairment and intellectual disability substantially affect communication, education, and independent daily functioning. Hearing loss, feeding difficulties, and motor/contracture involvement add to care burden. No formal EQ-5D/SF-36/PROMIS data exist for this ultra-rare disorder.

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4. Genetic / Molecular Information

Causal gene: ***MN1*** (Meningioma-1), chromosome **22q12.1**, HGNC:7180. *MN1* has an unusual **two-exon structure**, and its coding sequence is largely within the terminal exon.

Pathogenic variant class and location. All reported CEBALID-causing variants are **truncating (nonsense or frameshift)** and cluster in the **terminal exon or the extreme 3′ region of exon 1**. This genomic location is central to pathogenesis: it causes **escape from nonsense-mediated mRNA decay (NMD)**, so a truncated protein is actually produced. Mak et al.: *"All mutations, including the recurrent variant p.Arg1295* observed in 8/21 probands, fall in the terminal exon or the extreme 3' region of exon 1, and are therefore predicted to result in escape from nonsense-mediated mRNA decay"* (PMID: 31834374); NMD-escape was confirmed in patient fibroblasts.

Recurrent variant. p.Arg1295* (c.3883C>T) is a recurrent hotspot, present in 8/21 probands in Mak et al.—consistent with a CpG-dinucleotide mutational hotspot, not a population founder effect.

Functional consequence — gain/change of function. Miyake et al. demonstrated that *"the deletion of the C-terminal region led to increased protein stability, an inhibitory effect on cell proliferation, and enhanced MN1 aggregation in nuclei compared to what occurred in the wild type, suggesting that a gain-of-function mechanism is involved in this disease"* (PMID: 31839203). Mutant MN1 showed **impaired binding to ZBTB24 and to RING1** (an E3 ubiquitin ligase), proposed to interfere with ubiquitin-mediated proteasomal degradation and thereby increase the amount of the stabilized mutant protein.

Variant classification (ACMG/AMP): De novo truncating *MN1* variants in the NMD-escape region are classified **pathogenic/likely pathogenic** (standard PVS1 loss-of-function logic must be applied cautiously because canonical LOF is *not* the mechanism—position-aware interpretation is essential). Importantly, **MN1 whole-gene deletions cause a milder/different phenotype**, underscoring that variant *position and type* drive the CEBALID phenotype.

Allele frequency: Absent from population databases (gnomAD); these are private, de novo variants.

Somatic vs germline: The CEBALID variants are **germline (de novo)**. Separately—and distinct from CEBALID—*MN1* is a known somatic oncogene in acute myeloid leukemia (overexpression and *MN1* fusions such as MN1-ETV6/MN1-TEL), but this involves full-length/N-terminal MN1 overexpression, a different mechanism unrelated to the germline C-terminal truncation syndrome.

Modifier genes / epigenetics / chromosomal abnormalities: No disease-specific modifier genes or epigenetic signatures have been reported for CEBALID. No recurrent chromosomal abnormalities are associated (contrast: 22q microdeletions involving *MN1* produce a distinct, milder presentation).

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5. Environmental Information

Not applicable. CEBALID is a purely genetic, de novo dominant malformation syndrome. No environmental factors, toxins, lifestyle factors, or infectious agents are implicated in its causation or triggering.

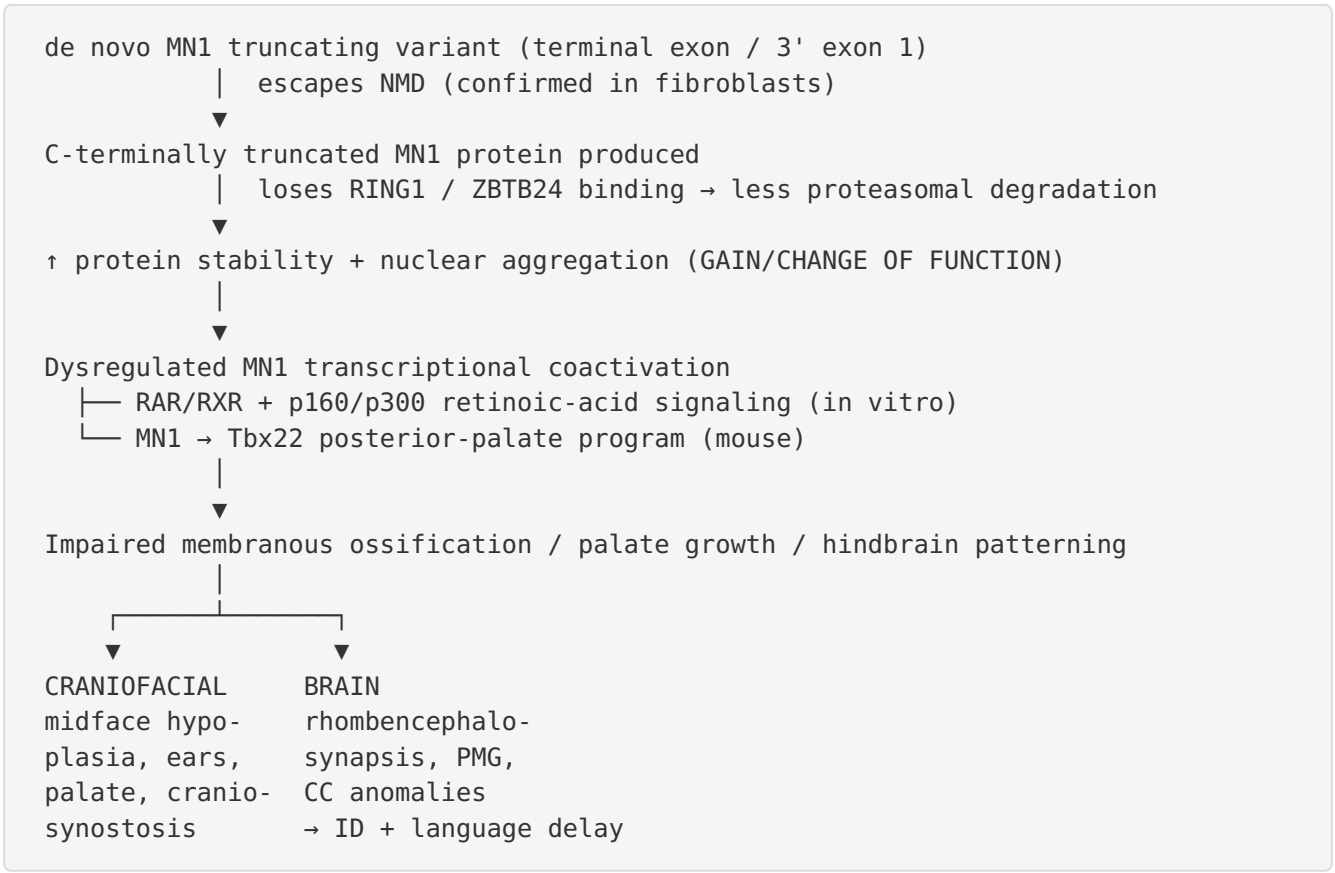
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6. Mechanism / Pathophysiology

Ordered causal chain

1. A **de novo truncating variant** (nonsense/frameshift) arises in the terminal exon / extreme 3′ of exon 1 of *MN1* → **leads to** a transcript with a premature stop codon positioned to **escape NMD** (demonstrated in patient fibroblasts).
2. NMD escape → **results in** translation of a **C-terminally truncated MN1 protein** (rather than no protein).

3. Loss of the C-terminus → **impairs binding to RING1 (E3 ubiquitin ligase) and ZBTB24 → leads to** reduced ubiquitin-mediated proteasomal degradation → **increased mutant protein stability** (demonstrated in vitro).
4. Stabilized mutant MN1 → **results in enhanced nuclear aggregation** and an **inhibitory effect on cell proliferation** (demonstrated in vitro) — a **gain/change-of-function** state, not haploinsufficiency.
5. Altered MN1 coactivator activity → **dysregulates MN1-dependent transcriptional programs**, including **retinoic-acid-receptor (RAR/RXR) signaling** (MN1 synergizes with p160/p300 coactivators) and the **MN1→*Tbx22* axis** in palate development (*inferred* link from model organisms to the human phenotype).
6. Disrupted transcriptional programs during embryogenesis → **impair intramembranous ossification of cranial bones and posterior palate growth** (demonstrated in *Mn1* mouse knockouts) and **perturb hindbrain/forebrain patterning → results in** the clinical malformations.
7. **Branch A (craniofacial):** → midface hypoplasia, dysmorphic ears, cleft/high-arched palate, craniosynostosis. **Branch B (brain):** → partial rhombencephalosynapsis, perisylvian polymicrogyria, corpus callosum anomalies → intellectual disability and severe expressive language delay.



Molecular pathways and protein function

MN1 is a **transcriptional coactivator**, not a sequence-specific transcription factor. van Wely et al. showed that *"MN1 is a transcription coactivator rather than a sequence-specific transcription factor, and that it may stimulate RAR/RXR-mediated transcription through interaction with p160 and p300"* and that *"The combination of MN1 and ligand-activated retinoic acid receptor leads to a synergistic induction of expression directed by the LTR"* (PMID: 12569362). This places MN1 within **retinoic acid / nuclear-receptor signaling** (RAR-RXR; CHEBI:26536 retinoic acid), a pathway central to craniofacial and neural development—offering a plausible mechanistic bridge to the CEBALID phenotype. MN1 additionally interacts with homeodomain transcription factors **PBX1 and PKNOX1** and with **ZBTB24**.

Developmental genetics of the palate

Liu et al. demonstrated that *"mice lacking the transcription factor Mn1 have defects in posterior but not anterior palatal growth"* and *"identified Tbx22, the mouse homolog of the human X-linked cleft palate gene, as a putative downstream target of Mn1 transcriptional activation"* (PMID:

18948418). MN1 activated human and mouse *Tbx22* promoters and dose-dependently increased endogenous *Tbx22* mRNA, defining an **MN1 → Tbx22** axis for posterior palate development.

Cranial bone development

Meester-Smoor et al. showed *Mn1*-null mice *"die at birth or shortly thereafter as the result of a cleft palate"* and concluded *"Mn1 is an important factor in development of membranous bones"* (PMID: 15870292). Null mice show agenesis of the alisphenoid, squamosal and vomer bones (formed almost exclusively by intramembranous ossification), with hypoplasia of frontal/parietal/interparietal bones—paralleling the human craniofacial skeletal phenotype.

Cell types and processes (ontology suggestions)

- **Cellular processes:** transcriptional coactivation (GO:0003713), RAR/RXR-mediated signaling, regulation of osteoblast differentiation (GO:0045667), intramembranous ossification (GO:0001957), cerebellum/hindbrain development (GO:0021549, GO:0030902), palate development (GO:0060021), protein stabilization/aggregation, ubiquitin-dependent protein catabolism (GO:0006511).
- **Cell types (CL):** osteoblast (CL:0000062), cranial neural crest cell (CL:0000333), cerebellar/hindbrain neuronal progenitors, palatal mesenchymal cells.
- **Subcellular (GO CC):** nucleus (GO:0005634) — site of mutant MN1 aggregation.

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7. Anatomical Structures Affected

Primary organs/systems: - **Brain / central nervous system** (UBERON:0000955) — cerebellum and vermis (UBERON:0002037 / UBERON:0004086) with rhombencephalosynapsis; cerebral cortex perisylvian region (polymicrogyria); corpus callosum (UBERON:0002336). - **Craniofacial skeleton** (UBERON:0010363 skeleton of skull) — midface, cranial membranous bones, palate (UBERON:0001716), orbits. - **Ears** (UBERON:0001690) — external ear dysmorphism; auditory apparatus (hearing loss).

Secondary/associated involvement: - **Musculoskeletal system** — joint contractures (arthrogryposis) in severe cases; hypotonia. - **Cardiopulmonary system** — poor neonatal adaptation in severe presentations (secondary). - **Cerebral vasculature** — persistent trigeminal artery.

Tissue/cell level: connective/skeletal tissue (membranous bone, osteoblasts), neural tissue (cerebellar and cortical neurons/progenitors), neural-crest-derived craniofacial mesenchyme.

Subcellular: nucleus (mutant protein aggregation).

Lateralization: Brain malformations are typically **midline/bilateral** (vermis, corpus callosum) or **bilateral** (perisylvian polymicrogyria); craniofacial features are **bilateral and symmetric**.

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8. Temporal Development

- **Onset: Congenital** (prenatal/neonatal). Structural malformations are established during embryogenesis; dysmorphism and brain anomalies are present at birth, with developmental delay apparent in infancy/early childhood.
- **Onset pattern:** Insidious with respect to neurodevelopment (delays recognized over time); malformations are present from birth.
- **Progression:** The disorder is **static/non-progressive (malformative)**. There is no neurodegeneration; disability is **lifelong and stable** rather than worsening.
- **Disease course:** Chronic, lifelong. No relapsing-remitting or episodic pattern. Severe neonatal presentations (e.g., arthrogryposis with cardiopulmonary maladaptation) carry higher early morbidity/mortality risk.
- **Critical period:** The structural/developmental window is **embryonic craniofacial and hindbrain morphogenesis** (already passed by birth), making structural prevention impossible post-natally; early-childhood developmental intervention is the actionable window for functional outcomes.

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9. Inheritance and Population

- **Inheritance pattern: Autosomal dominant**, essentially always **de novo**. Mak et al.: *"We identified a cluster of de novo truncating mutations in MN1"* (PMID: 31834374). Recurrence risk for parents of an affected child is low (apart from rare germline mosaicism).
- **Penetrance:** Appears **complete** for the recognizable gestalt among carriers of the specific C-terminal truncating variants.

- **Expressivity: Variable**, ranging from the "classic" gestalt to severe neonatal arthrogryposis presentations.
- **Genetic anticipation:** Not applicable (not a repeat-expansion disorder).
- **Germline mosaicism:** Theoretically possible but not well documented; relevant for rare recurrence.
- **Founder effects / consanguinity:** None. The recurrent **p.Arg1295*** reflects a **CpG mutational hotspot**, not a founder allele; consanguinity is irrelevant for a de novo dominant disorder.
- **Carrier frequency:** Not applicable (de novo, not inherited).

Epidemiology / demographics: - **Prevalence/incidence:** Unknown—ultra-rare. Dirani et al.: *"less than 30 cases have been reported worldwide"* (PMID: 40713540). - **Sex ratio:** No established sex predilection. - **Ethnic/geographic distribution:** No established predilection; cases reported across multiple populations.

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10. Diagnostics

Recommended approach: Clinical suspicion from the gestalt (**midface hypoplasia + severe expressive language delay + partial rhombencephalosynapsis on MRI**), confirmed by **molecular detection of a heterozygous truncating *MN1* variant in the terminal exon or extreme 3′ of exon 1**.

Genetic testing: - **Whole-exome sequencing (WES)** — the principal confirmatory test. Dirani et al.: *"Whole exome sequencing confirmed the diagnosis of CEBALID syndrome"* (PMID: 40713540). - **Whole-genome sequencing (WGS)** — also effective. - **Single-gene / targeted *MN1* sequencing** — appropriate when the phenotype is strongly suggestive. - **Interpretation caveat:** Variant **position and type are decisive**—3′ truncating (NMD-escape) variants cause CEBALID, whereas **whole-gene deletions produce a different, milder phenotype**. A chromosomal microarray detecting an *MN1* deletion should *not* be equated with CEBALID.

Imaging: - **Brain MRI** is the key diagnostic adjunct, demonstrating **partial rhombencephalosynapsis** (vermian hypoplasia with cerebellar hemisphere fusion), **perisylvian polymicrogyria**, corpus callosum anomalies, abnormal posterior clinoid processes, and persistent trigeminal artery.

Other clinical tests: - **Audiologic evaluation** (conductive/sensorineural hearing loss). - **Ophthalmologic assessment** (shallow orbits/proptosis). - No specific biochemical biomarker or metabolic test exists.

Differential diagnosis: Other causes of rhombencephalosynapsis (e.g., Gómez-López-Hernández syndrome, VACTERL-H), other midface-hypoplasia syndromes, and other polymicrogyria syndromes. The specific combination of midface hypoplasia, severe expressive language delay, and partial rhombencephalosynapsis plus a 3′ truncating *MN1* variant is distinguishing.

Screening: No population or newborn screening exists (ultra-rare, de novo). Cascade testing of parents is generally uninformative given the de novo origin but may be offered for recurrence-risk counseling and mosaicism assessment.

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11. Outcome / Prognosis

- **Survival/mortality:** No formal survival statistics exist. The disorder is **not inherently lethal** in classic presentations; most reported individuals survive with lifelong disability. **Severe neonatal presentations** (arthrogryposis with cardiopulmonary maladaptation requiring mechanical ventilation) carry elevated early morbidity and potential mortality.
- **Morbidity/function:** Substantial, driven by **intellectual disability and severe expressive language impairment**, with additional impact from hearing loss, feeding difficulties, motor involvement, and craniofacial anomalies.
- **Disease course:** Static/non-progressive; disability is stable and lifelong.
- **Complications:** Feeding difficulties, hearing impairment, sequelae of cleft palate/craniosynostosis, and in severe cases respiratory compromise.
- **Prognostic factors:** Severity of brain malformation and neonatal cardiopulmonary/contracture involvement likely predict outcome, though no validated prognostic model exists for this small case set.

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12. Treatment

There is no disease-specific or curative pharmacotherapy. Management is **supportive and multidisciplinary**, targeting individual manifestations:

Domain	Intervention	NCIT concept (suggested)
Developmental	Early intervention, special education	Developmental therapy
Communication	Speech/language therapy (severe expressive delay)	Speech Therapy (NCIT:C15336)
Motor	Physical & occupational therapy (hypotonia, contractures)	Physical Therapy (NCIT:C15368); Occupational Therapy (NCIT:C15337)
Hearing	Audiologic evaluation, hearing aids	Audiologic assessment
Vision	Ophthalmologic monitoring	Ophthalmologic examination
Craniofacial	Surgical repair of cleft palate; craniosynostosis surgery	Surgical Procedure (NCIT:C15329)
Feeding	Feeding support / nutritional management	Nutritional Support (NCIT:C15335)
Neonatal (severe)	Respiratory support/mechanical ventilation as needed	Mechanical Ventilation (NCIT:C70902)
Family	Genetic counseling	Genetic Counseling (NCIT:C15648)

- **Pharmacogenomics / targeted / gene / cell / RNA therapies:** None currently applicable or in trials for CEBALID.
- **Experimental treatments:** No disease-specific clinical trials (no NCT identifiers) are available.
- **Personalized medicine:** Limited to position-aware genetic diagnosis for counseling and anticipatory care.

13. Prevention

- **Primary prevention:** Not possible—de novo variant; no modifiable risk factors.
- **Secondary prevention:** Early developmental intervention and anticipatory management of hearing, feeding, vision, and craniofacial complications to optimize functional outcomes.
- **Tertiary prevention:** Multidisciplinary management of complications (cleft palate repair, craniosynostosis surgery, hearing/feeding support).
- **Genetic counseling:** Essential. Because variants are de novo, **recurrence risk for parents is low** (rare germline mosaicism excepted). Prenatal/preimplantation testing is feasible for a known familial variant but rarely indicated given the low recurrence risk.
- **Immunization / public health / behavioral / environmental interventions:** Not applicable.

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14. Other Species / Natural Disease

- **Orthologous gene:** Mouse ***Mn1*** (human *MN1* NCBI Gene ID 4330). The gene and its developmental roles are **evolutionarily conserved**.
- **Natural disease in other species:** No naturally occurring CEBALID-equivalent disease is documented in companion animals or wildlife (no OMIA entry for a spontaneous *MN1* truncation disorder). The animal evidence comes from **engineered mouse models** rather than natural disease.
- **Comparative pathology:** *Mn1*-null mice recapitulate the **craniofacial/palate** component (cleft palate, membranous bone agenesis) but model loss of function rather than the human gain-of-function truncation—an important caveat for cross-species comparison.
- **Zoonotic/transmission:** Not applicable (genetic disorder).

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15. Model Organisms

- **Mouse (*Mus musculus*): *The principal model. Constitutive Mn1 knockout mice (Meester-Smoor et al.) die at birth from cleft palate and show agenesis/hypoplasia of intramembranous cranial bones; heterozygotes show hypoplastic membranous bones***

with incomplete-penetrance cleft palate, and Mn1+/− animals followed to 30 months showed no increased tumor incidence (PMID: 15870292). These models established MN1's role in membranous bone and palate development and the downstream MN1→Tbx22 axis (PMID: 18948418).

- **In vitro / cellular models:** Transfection assays (e.g., NIH3T3) demonstrated **MN1 coactivation of RAR/RXR via p160/p300** (PMID: 12569362) and, for the disease variants, **increased protein stability, nuclear aggregation, proliferation inhibition, and altered interactor binding (RING1, ZBTB24, PBX1, PKNOX1)** (PMID: 31839203). Patient-derived **fibroblasts** confirmed NMD escape of the mutant transcript.
- **Phenotype recapitulation / limitations:** Mouse *Mn1* knockouts reproduce the **craniofacial/palatal** phenotype well but model **loss of function**, whereas human CEBALID is a **gain/change-of-function** truncation. **No mouse model expressing the human C-terminal truncation has been reported**, and the **brain malformation (rhombencephalosynapsis)** is not captured by existing knockouts—key gaps for mechanistic study.

Key Findings (Consolidated)

#	Finding	Primary evidence (PMID)
F1	CEBALID is caused by de novo C-terminal truncating *MN1* variants acting via gain-of-function	31839203, 31834374
F2	Variants escape NMD; recurrent p.Arg1295* in 8/21 probands; truncated protein produced	31834374
F3	Truncated MN1 shows ↑stability, nuclear aggregation, ↓proliferation, impaired RING1/ZBTB24 binding	31839203
F4	Hallmark gestalt: midface hypoplasia + severe expressive language delay + ID	31834374
F5	Distinctive brain malformation: partial rhombencephalosynapsis (8/10 imaged)	31834374, 40713540
F6	MN1 is a RAR/RXR transcriptional coactivator synergizing with p160/p300	12569362
F7	MN1 acts upstream of *Tbx22*, regulating posterior palate growth	18948418
F8	MN1 essential for intramembranous ossification of cranial bones (mouse null → cleft palate)	15870292
F9	Ultra-rare (<30 cases); AD de novo; WES-confirmed; supportive management	40713540, 31834374

Mechanistic Model / Interpretation (Synthesis)

CEBALID is best understood as a **dominant "toxic/altered-product" disorder of a developmental transcriptional coactivator**. The unusual two-exon architecture of *MN1* means that truncating variants in the terminal exon escape NMD and produce a stable, C-terminally truncated protein. Loss of the C-terminus disrupts interactions (RING1, ZBTB24) that normally promote MN1's proteasomal turnover, so the aberrant protein **accumulates and aggregates in the nucleus**, producing a gain/change-of-function rather than simple loss.

Because wild-type MN1 normally **coactivates retinoic-acid-receptor signaling** and drives developmental programs including the **MN1→Tbx22** posterior-palate axis and **intramembranous ossification**, the dysregulated mutant perturbs **craniofacial morphogenesis** (midface hypoplasia, ears, palate, cranial bones) and **hindbrain/forebrain patterning** (rhombencephalosynapsis, polymicrogyria), yielding the paired craniofacial–brain phenotype and its neurodevelopmental consequences. The strongest-evidence links are the human genetic/functional data (Mak, Miyake) and the mouse craniofacial data (Meester-Smoor, Liu); the retinoic-acid and *Tbx22* connections to the full human brain phenotype remain **plausible but partly inferred**.

Evidence Base

PMID	Title (abbrev.)	Evidence type	Contribution
31834374	*MN1 C-terminal truncation syndrome ... with partial rhombencephalosynapsis* (Mak 2020)	Human clinical (n=23)	Defines phenotype gestalt, recurrent p.Arg1295*, NMD-escape mechanism, rhombencephalosynapsis
31839203	*Gain-of-Function MN1 Truncation Variants...* (Miyake 2020)	Human + in vitro (n=3)	Establishes de novo causation and gain-of-function mechanism (↑ stability, aggregation)
40713540	*Extremely rare case of CEBALID presenting as congenital arthrogryposis* (Dirani 2025)	Human case report	CEBALID acronym, rarity (<30 cases), WES diagnosis, severe neonatal/arthrogryposis spectrum
15870292	*Targeted disruption of Mn1... membranous bones* (Meester-Smoor 2005)	Mouse model	MN1 essential for membranous bone & palate development
18948418	*Mn1 acts upstream of Tbx22...* (Liu 2008)	Mouse model	MN1→Tbx22 axis for posterior palate growth
12569362	*MN1 oncoprotein synergizes with RAC3 and p300...* (van Wely 2003)	In vitro	MN1 is RAR/RXR coactivator via p160/p300

Context literature (not CEBALID-causal): Multiple papers (PMID: 33542482, 26927674, 22905229, 22488406, 23646898, 19379599) document *MN1*'s **somatic oncogenic role in AML** (overexpression, MN1 fusions, poor prognosis). These are mechanistically distinct from the germline C-terminal truncation syndrome but reinforce MN1's biology as a dosage-sensitive transcriptional coactivator.

Limitations and Knowledge Gaps

1. **Tiny evidence base:** Fewer than ~30 reported cases; all epidemiology (prevalence, incidence, sex/ethnic distribution, survival) is unavailable or estimated.
2. **No humanized model:** No mouse or organoid model expressing the human C-terminal truncation; the hallmark **rhombencephalosynapsis** is not recapitulated by existing *Mn1* knockouts.
3. **Mechanistic inference:** The RAR/RXR-coactivation and *Tbx22* pathways are established for wild-type MN1 and palate, but their precise disruption by the truncated protein—and their link to the human brain malformation—remains partly inferred.
4. **Genotype–phenotype detail:** Why some variants (e.g., arthrogryposis presentations) are more severe is unknown; no validated modifier genes.
5. **No biomarkers or targeted therapy:** No molecular diagnostic biomarker beyond sequencing; no disease-specific treatment or clinical trials.

Proposed Follow-up Experiments / Actions

1. **Generate a knock-in mouse (or human iPSC-derived organoid)** carrying the recurrent **p.Arg1295*** truncation to test whether the stabilized mutant produces rhombencephalosynapsis and midface hypoplasia—directly modeling gain-of-function.
2. **Cerebellar/cortical organoids** from patient iPSCs to dissect hindbrain patterning defects and screen for pathway rescue (e.g., retinoic-acid signaling modulators).
3. **Transcriptomic/ChIP profiling** of mutant vs wild-type MN1 in neural-crest and neural progenitor cells to define dysregulated target genes (test *TBX22* and RAR/RXR targets).
4. **Establish an international CEBALID registry** to capture natural history, genotype–phenotype correlations, and outcomes across the <30 known cases and future diagnoses.

5. **Degradation-targeting proof of concept:** Since the pathogenic protein is abnormally stabilized, test whether restoring its proteasomal turnover (e.g., modulating RING1-mediated ubiquitination) reduces nuclear aggregation in cellular models.

Report compiled from 8 confirmed findings across 5 investigation iterations and 13 reviewed papers. Evidence types span human clinical case series, mouse genetic models, and in vitro functional assays.
